

Welcome to the October 2026 Spatial Transcriptomics Training Workshop!

WIFI: ALSF - Guest
Password: MakeADifference

October 6–8, 2026
Childhood Cancer Data Lab

<https://alexslemonade.github.io/2026-october-training>

Childhood Cancer
Data  **Lab** **x**



Tell us about you!

- What's your name?
- Where do you work?
- What are you studying?
- What is a recent piece of media (book, tv, movie, live performance...) you enjoyed?



Meet your instructors



JOSH

Joshua Shapiro

Senior Data Scientist @ the Data Lab

PhD Ecology & Evolution, *UChicago*

Postdoc Integrative Genomics, *Princeton*

Research interests:

- **Evolutionary genomics**
- **Single cell workflows**



jashapiro

Meet your instructors



STEPHANIE

Stephanie Spielman

former Data Scientist @ the Data Lab

PhD Integrative Biology *UT Austin*

Postdoc Computational Molecular Evolution *Temple*

Research interests:

- **Protein evolution and comparative sequence analysis**
- **Data science and bioinformatics education**



sjspielman

Meet your instructors



ALLY

Ally Hawkins

Data Science Manager @ the Data Lab

PhD Cancer Biology *University of Michigan*

Postdoc Computational Biology *Cornell*

Research interests:

- **Single cell data analysis**
- **Origins of pediatric solid tumors**

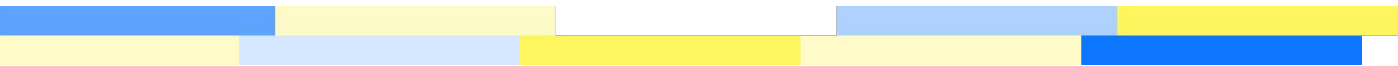


allyhawkins

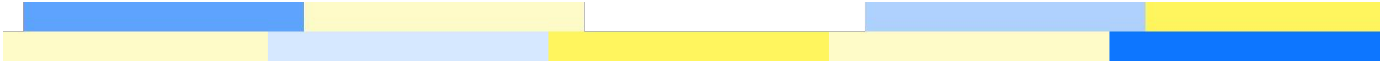
Tell us about you!

- What's your name?
- Where do you work?
- What are you studying?
- What is a recent piece of media (book, tv, movie, live performance...) you enjoyed?





Code of Conduct



Be kind, have fun

We value the involvement of everyone in the community. We are committed to creating a friendly and respectful place for learning, teaching, and contributing.

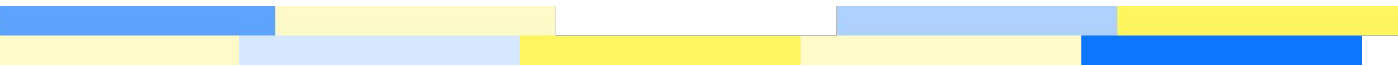
- Use welcoming and inclusive language
- Be respectful of different viewpoints and experiences
- Gracefully accept constructive criticism
- Focus on what is best for the community
- Show courtesy and respect towards other community members

Read the full Code of Conduct here:

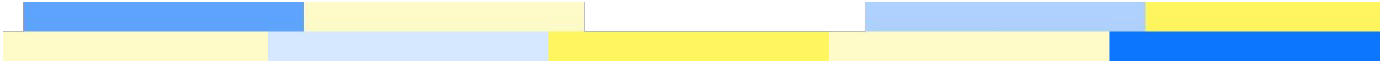
<https://alexslemonade.github.io/2026-october-training/code-of-conduct.html>



If you at any time feel harassed or treated inappropriately, please contact
ccd1@alexslemonade.org



What you will learn (and what you won't)



What you will learn

- How to import & QC spatial transcriptomics data in R with **SpatialExperiment** objects
 - We will discuss 10x Visium and Xenium platforms only
- How to incorporate spatial data in clustering
- How to perform cell type deconvolution
- Methods for identification of spatially variable genes
- Spatial neighborhood analysis



What you won't learn

- We won't address experimental design (e.g., how many replicates you need).
 - We won't cover every spatial technology; the field is moving fast!
 - We won't cover pre-processing of spatial data (alignment, quantification, and cell segmentation).
 - We won't cover every feature or assumption of the tools we do present, but we will try to highlight the features and gotchas that we think are relevant to most users.
 - We won't compare packages and algorithms much, but we will try to point you to places where you can learn more about comparisons.
 - You may not be able to perform every analysis you need for your own work, particularly for complex experimental designs.
- 

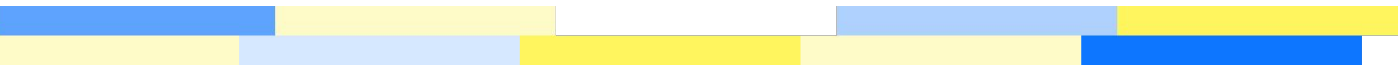
How do we pick what we teach?

We want methods to be or to have:

- Useful for a wide range of experimental designs, sample sizes
- Easy to use, well-documented, and consistently updated
- Solid tutorials, a sizeable user base, and responsive authors/maintainers

We have a preference for methods that integrate easily into a single workflow that can be run on a laptop (and our own personal biases as scientists).





Schedule



Tuesday

Workshop Intro

Modules

- Introduction to Spatial Transcriptomics
- Spatial Clustering

Consultations

- Exercise notebooks

Wednesday

Modules

- Deconvolution
- Spatially Variable Genes

Consultations

- Exercise notebooks
- Your own data

Thursday

Modules

- Processing Xenium Data
- Spatial Neighborhood Analysis

Consultations

- Exercise notebooks
- Your own data

Presentations

Daily Schedule Outline

Instruction

Full group

Lectures

- Introduce concepts and background
- Demonstrate usage
- Answer general questions

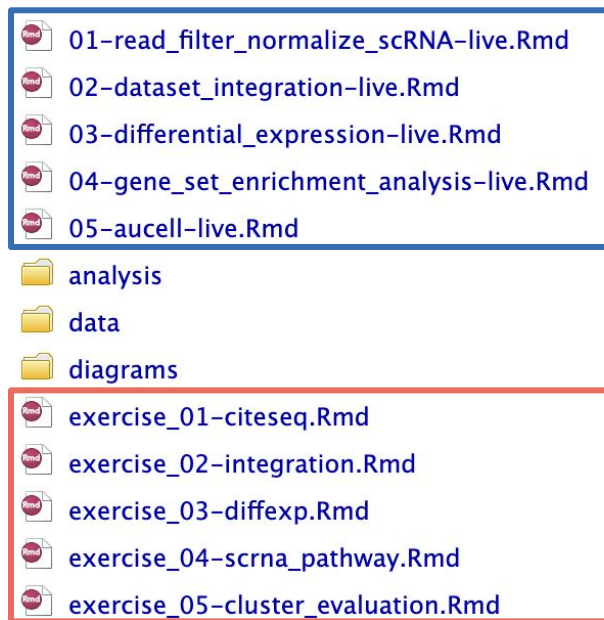
Consultation Period

Exercise notebooks

Your own data

- Ask questions of instructors and other participants
- Practice what you have learned
- Work on exercises individually or in groups
- Work with your own data

Module Layout



These are **Instruction** notebooks.
We'll walk through these together,
adding code during the workshop.

These are **Exercise** notebooks.
Use these to practice what you have
learned. We're here to help!

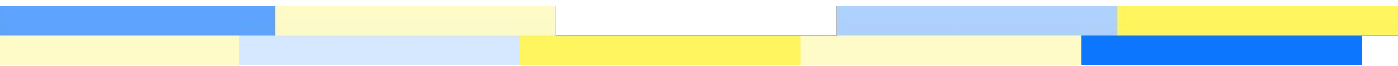
Thursday Afternoon

Your own projects Exercise notebooks

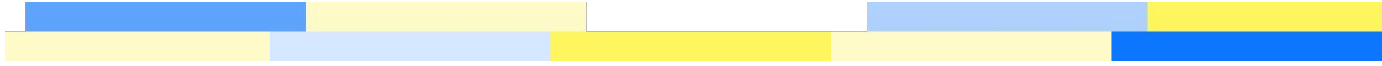
Spend Thursday afternoon working with your own data, getting assistance as needed from Data Lab staff and each other.

Presentations

Present what you worked on during the consultation times to the group!



Training Procedures



We're going to use sticky notes and note cards...

- As an alternative to raising your hands for help
- To give feedback about the session



During the session

Use the yellow sticky note to let us know that you need help or having some trouble following along.

One of our helpers will come over and help you.



Use the blue sticky note to let us know when a long-running step has completed.



At the end of each session

At the end of each module,
write down your muddiest point on
a white note card:

I do not understand
_____.

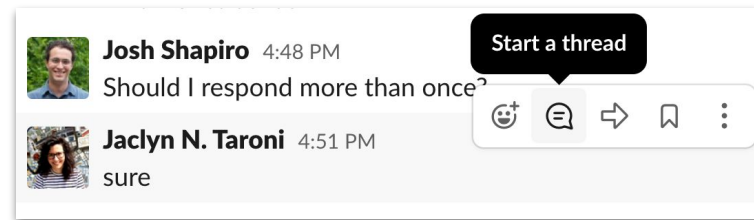
At the end of the session, use one
blue note card and one white note
card to write down:

What did you like about this
session?

How might we improve the
session?

We encourage you to use Slack

- You have been added to the **#2026-october-training** Slack channel
- Post public questions, get help with errors and debugging, make comments, and help others!
 - Use threads to keep related content together
- Stay in touch after the workshop!



Housekeeping Notes

- Waivers – If you have not yet done so, we have hard copies of waivers available.
- Where are the restrooms?
- Where is water/coffee available?
- Snacks!
- Dinner tomorrow (Wednesday) at 6:00PM at [SOMO Manayunk](#)
4311 Main Street, Philadelphia, PA 19127